Create daemon.sh

```
priyansh@IDEAPAD-18UT:~$ nano daemon.sh
priyansh@IDEAPAD-18UT:~$
```

Daemon.sh – script

```
  GNU nano 7.2                        daemon.sh
#!/bin/bash

while true
do
  echo "Daemon running at $(date)" >> daemon.log
  sleep 5
done

                               [ Read 7 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify
```

Ctrl + S – save it & Ctrl + X – exit

Make script executable

```
priyansh@IDEAPAD-18UT:~$ chmod +x daemon.sh
```

Run the script in Background - ./daemon.sh &
Check on the process - ps aux | grep daemon.sh

```
priyansh@IDEAPAD-18UT:~$ ./daemon.sh &
[1] 453
priyansh@IDEAPAD-18UT:~$ ps aux | grep daemon.sh
priyansh     453  0.0  0.0   4752  3328 pts/0    S    22:50   0:00 /bin/bash
 ./daemon.sh
priyansh     459  0.0  0.0   4096  1920 pts/0    S+   22:50   0:00 grep --co
lor=auto daemon.sh
```

Kill the process using PID

```
priyansh@IDEAPAD-18UT:~$ kill 453
priyansh@IDEAPAD-18UT:~$ kill 459
-bash: kill: (459) - No such process
[1]+  Terminated              ./daemon.sh
priyansh@IDEAPAD-18UT:~$
```

Create service file -

Type the password and you'll enter script edit mode

```
priyansh@IDEAPAD-18UT:~$ sudo nano /etc/systemd/system/mydaemon.service
[sudo] password for priyansh:
priyansh@IDEAPAD-18UT:~$
```

mydaemon.sh – script

```
Ubuntu
  GNU nano 7.2          /etc/systemd/system/mydaemon.service
[Unit]
Description=My Daemon Service

[Service]
ExecStart=/home/$USER/daemon.sh
Restart=always

[Install]
WantedBy=multi-user.target
```

Enable and Start the services

```
priyansh@IDEAPAD-18UT:~$ sudo systemctl daemon-reexec
sudo systemctl daemon-reload
sudo systemctl enable mydaemon
sudo systemctl start mydaemon
Created symlink /etc/systemd/system/multi-user.target.wants/mydaemon.service
 → /etc/systemd/system/mydaemon.service.
priyansh@IDEAPAD-18UT:~$ systemctl status mydaemon
```

```
× mydaemon.service - My Daemon Service
     Loaded: loaded (/etc/systemd/system/mydaemon.service; enabled; preset: enabled)
     Active: failed (Result: exit-code) since Thu 2026-04-16 22:56:00 UTC; 1min 6s ago
   Duration: 3ms
    Process: 627 ExecStart=/home/$USER/daemon.sh (code=exited, status=203/EXEC)
   Main PID: 627 (code=exited, status=203/EXEC)
        CPU: 2ms

Apr 16 22:56:00 IDEAPAD-18UT systemd[1]: mydaemon.service: Scheduled restart job, restart counter is at 5.
Apr 16 22:56:00 IDEAPAD-18UT systemd[1]: mydaemon.service: Start request repeated too quickly.
Apr 16 22:56:00 IDEAPAD-18UT systemd[1]: mydaemon.service: Failed with result 'exit-code'.
Apr 16 22:56:00 IDEAPAD-18UT systemd[1]: Failed to start mydaemon.service - My Daemon Service.
```

Create CronTab

```
priyansh@IDEAPAD-18UT:~$ crontab -e
no crontab for priyansh - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]: 1
crontab: installing new crontab
priyansh@IDEAPAD-18UT:~$
```

Crontab – script

```
Ubuntu
 GNU nano 7.2                          /tmp/crontab.qBdq3H/crontab
* * * * * /home/$USER/daemon.sh
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
```

Crontab l- to check

```
priyansh@IDEAPAD-18UT:~$ crontab -l
* * * * * /home/$USER/daemon.sh
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
priyansh@IDEAPAD-18UT:~$
```

To Install Fuse -

sudo apt update

sudo apt install fuse3 python3-pip -y

pip3 install fusepy

(It can take some time to install Fuse)

```
Ubuntu
priyansh@IDEAPAD-18UT:~$ sudo apt update
sudo apt install fuse3 python3-pip -y
pip3 install fusepy
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1608 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1902 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [257 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.8 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1169 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [345 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1666 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2822 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2957 kB]
Get:23 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [658 kB]
Get:24 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:25 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [7204 B]
Get:26 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [689 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.5 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [8208 B]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [512 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7368 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1484 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 15.8 MB in 15s (1063 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
113 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Create fuse_fs.py

```
priyansh@IDEAPAD-18UT:~$ nano fuse_fs.py
priyansh@IDEAPAD-18UT:~$
```

Fuse_fs.py - script

```
Ubuntu

GNU nano 7.2
from fuse import FUSE, Operations

class SimpleFS(Operations):
    def readdir(self, path, fh):
        return ['.', '..', 'file.txt']

    def read(self, path, size, offset, fh):
        return b"Hello from FUSE filesystem\n"

if __name__ == '__main__':
    FUSE(SimpleFS(), 'mnt', foreground=True)
```

Create mount directory and then run the FUSE Filesystem

```
priyansh@IDEAPAD-18UT:~$ nano fuse_fs.py
priyansh@IDEAPAD-18UT:~$ mkdir mnt
priyansh@IDEAPAD-18UT:~$ python3 fuse_fs.py
```

The Terminal will get busy for this so open a fresh one and then type these to get the output -
ls mnt
cat mnt/file.txt - "Hello from FUSE Filesystem"
And then unmount FUSE - fusermount -u mnt

```
priyansh@IDEAPAD-18UT:~$ ls mnt
cat mnt/file.txt
```